Dyadic Brackets and Density-One Descent in the Collatz Map

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Abstract

We study the Collatz map via the fully accelerated transformation

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

which maps odd integers to odd integers by removing all powers of two generated at each odd step.

Finite trajectories are encoded by *division words*, recording the sequence of 2-adic valuations encountered along an orbit. Each division word induces an exact affine transformation together with an associated congruence condition on admissible initial values. This yields a symbolic–affine framework in which admissible words correspond to residue classes modulo powers of two.

These residue classes organize naturally into a forest of nested cylinders. Contractive division words define first-passage cylinders that remove positive measure from the residual set, while non-contractive extensions form a shrinking survivor forest.

The symbolic growth of admissible division words is governed by an induced grammar whose combinatorial expansion is controlled by Sturmian return words associated with the continued-fraction expansion of $\log_2(3)$. We show that the resulting cocycle growth remains strictly subcritical: the number of admissible words grows like ρ^m with $\rho < 3$, while the associated congruence moduli grow at a rate comparable to powers of 3 and therefore refine more rapidly than symbolic proliferation can compensate.

This creates a competition between symbolic growth and arithmetic refinement that forces exponential sparsification of the surviving forest. Furthermore, the number of contractive cylinders required to cover successive dyadic

scales remains bounded, producing a stable dyadic covering mechanism whose cumulative contribution approaches full measure.

As a consequence, the set of integers that fail to admit an eventual descent prefix has natural density zero. Equivalently, the set of integers that eventually decrease under iteration of the Collatz map possesses natural density one.

This work does not exclude the existence of exceptional trajectories. Rather, it shows that any such exceptions must lie within a residual survivor set of density zero and cannot arise from any scalable arithmetic structure generated by the affine grammar.

Contents

1 Introduction

The $3x+1$ problem, also known as the Collatz conjecture [1], concerns the behavior of the map

$$T(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ 3x+1, & \text{if } x \text{ is odd.} \end{cases}$$

on the positive integers. It asserts that every positive integer eventually reaches 1 under iteration of T. Despite its elementary formulation, the conjecture has resisted proof for decades [2, 3]. Extensive computational verification confirms convergence for very large ranges [4], but does not reveal the global structure governing trajectory behavior.

A major advance by Tao [5] established that almost all integers eventually attain values smaller than their initial value under iteration of the Collatz map in the sense of logarithmic density.

The present work develops a deterministic, arithmetic framework for analyzing typical Collatz behavior based on a symbolic–affine encoding of trajectories. Using this approach, we encode the dynamics symbolically rather than studying individual trajectories with the fully accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

which maps odd integers to odd integers by compressing each odd step into a single transformation.

Each trajectory under f is represented by a finite *division word*, recording the number of factors of 2 removed (the 2-adic valuation) at each iterate. These words admit an exact affine representation obtained by symbolic composition of the accelerated map: each division word induces a map of the form

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

and determines a corresponding congruence class modulo $2^{D(\omega)}$. Thus division words determine canonical arithmetic *constraint classes* (residue classes) in the integers, once realizability is imposed.

This correspondence allows the dynamics to be studied at the level of residue classes rather than individual orbits. Convergent division words—those for which the associated affine map is contractive—identify families of integers that decrease after a fixed number of steps. The global problem is thereby reduced to understanding how these families are distributed and how their densities accumulate.

Two competing effects govern this distribution. On one hand, the number of admissible convergent division words grows exponentially with length. On the other hand, each word imposes a congruence constraint whose modulus grows exponentially with the total 2-adic valuation. Using structural constraints on admissible division words, encoded by an increment grammar, we show that the combinatorial growth rate is strictly subcritical: the number of admissible words grows like ρ^m for some $\rho < 3$, while the moduli scale like powers of 2 satisfying $3^m < 2^{D(\omega)} < 2 \cdot 3^m$.

Admissible division words organize naturally into a forest of nested cylinders. Contractive words define first-passage cylinders that remove positive measure from the residual set, while non-contractive extensions form a shrinking survivor forest.

This transforms the Collatz problem into a competition between symbolic growth and arithmetic refinement, with convergence governed by their relative exponential rates. The resulting arithmetic refinement is governed by the Sturmian structure induced by the continued-fraction expansion of $\log_2(3)$, whose return words generate a hierarchy of cocycle towers controlling symbolic growth.

This imbalance leads to exponential sparsification of the residual forest. Contractive cylinders remove positive dyadic mass at every scale, while the number of cylinders required to cover successive dyadic intervals remains uniformly bounded. Consequently the residual mass cannot replenish itself faster than contractive cylinders remove it.

Empirically, the interaction between symbolic growth and arithmetic refinement produces a stable dyadic covering plateau, suggesting that bounded collections of contractive cylinders continue to cover successive scales indefinitely.

The objective of this work is therefore not to prove global convergence of the Collatz map, but rather to demonstrate that the set of integers admitting eventual descent possesses natural density one.

While this does not exclude the existence of exceptional trajectories, any such exceptions must lie within a set of zero density and cannot arise from any scalable arithmetic structure generated by the symbolic dynamics. Any counterexample to the Collatz conjecture, if it exists, must therefore avoid every contractive cylinder generated by the affine grammar and must persist indefinitely within the residual survivor forest.

Structure of the Paper. The argument proceeds as follows:

- Section 2 introduces division words and their total division counts.
- Section 3 develops the affine representation of division words and their congruence structure.
- Section 4 establishes admissibility and minimal valuation constraints.
- Section 5 establishes the stepwise realizability of admissible division words and the unique residue lifting property.
- Section 6 defines the geometry of contractive cylinders.
- Section 7 explores the branching structure of the affine grammar.
- Section 8 discusses competition between symbolic entropy and dyadic growth.
- Section 9 establishes the collapse of measure for disjoint cylinders.
- Section 10 establishes extremal growth properties with Sturmian return words.

2 Symbolic Encoding of Collatz Trajectories

2.1 Division Words

Definition 2.1 (Division Word)**.** A division word of length $m \geq 1$ is a sequence

$$\omega = (d_0, \dots, d_{m-1}),$$

where each $d_i \in \mathbb{N}$ denotes the number of factors of 2 removed at the i-th iterate of the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Remark. Since f maps odd integers to odd integers, each $3f^i(x)+1$ is even, so $d_i \geq 1$ for all i.

Definition 2.2 (Realization)**.** An odd integer $x \in \mathbb{N}$ realizes ω if

$$d_i = v_2(3f^i(x) + 1), \quad 0 \leq i < m.$$

Remark. Every odd integer x determines a unique infinite valuation sequence

$$(v_2(3x+1),\, v_2(3f(x)+1),\, v_2(3f^2(x)+1), \dots),$$

of which a division word is a finite prefix.

2.2 Total Division Count

Definition 2.3. The total division count of a word ω is

$$D(\omega) := \sum_{i=0}^{m-1} d_i.$$

Remark. The quantity $D(\omega)$ measures the cumulative dyadic contraction along the symbolic trajectory encoded by ω.

3 Affine Representation of Division Words

Definition 3.1. Each division word ω of length m determines a formal affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

obtained by symbolic composition of the accelerated Collatz map.

Proposition 3.1 (Concatenation and Nested Affine Structure)**.** Let ω and η be division words. If the concatenation $\omega\eta$ denotes the word obtained by appending η to ω, then the associated affine maps satisfy

$$F_{\omega\eta} = F_\eta \circ F_\omega.$$

Consequently, division-word concatenation induces a semigroup homomorphism from the free semigroup of division words under concatenation to the semigroup of affine transformations under composition.

Furthermore, if C_ω denotes the set of integers realizing the word ω, then

$$C_{\omega\eta} \subseteq C_\omega.$$

Thus extensions of division words correspond to nested arithmetic cylinders.

Proof. Write

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

and similarly

$$F_\eta(x) = \frac{3^n x + c(\eta)}{2^{D(\eta)}}.$$

Then

$$F_\eta(F_\omega(x)) = \frac{3^n \left(\frac{3^m x + c(\omega)}{2^{D(\omega)}}\right) + c(\eta)}{2^{D(\eta)}}.$$

Rearranging gives

$$F_\eta(F_\omega(x)) = \frac{3^{m+n} x + 3^n c(\omega) + 2^{D(\omega)} c(\eta)}{2^{D(\omega)+D(\eta)}}.$$

The denominator exponent and ternary exponent add under concatenation, while the affine constant combines according to the usual affine composition rule. Therefore this map is precisely the affine representation associated with the concatenated word $\omega\eta$.

For the cylinder inclusion, any integer realizing $\omega\eta$ must first realize the prefix ω before realizing the appended word η. Hence every realization of $\omega\eta$ is also a realization of ω, and therefore

$$C_{\omega\eta} \subseteq C_\omega.$$

□

Definition 3.2 (Contractive and Expansive Words)**.** Let ω be a realizable division word of length m.

The word ω is called:

- contractive if
$$F_\omega(x) < x$$
for every realization of ω;
- expansive if
$$F_\omega(x) \geq x$$
for every realization of ω.

Remark. Contractivity in this sense establishes only finite descent:

$$f^m(x) < x.$$

It does not imply that subsequent iteration converges to 1.

3.1 Dyadic-Ternary Balance

The symbolic dynamics of a division word are governed by the balance between the cumulative ternary growth 3^m and the cumulative dyadic contraction $2^{D(\omega)}$.

For a division word ω of length m, the associated affine map has linear coefficient

$$\frac{3^m}{2^{D(\omega)}}.$$

Thus:

- if $2^{D(\omega)} < 3^m$, the symbolic trajectory is linearly expansive,
- if $2^{D(\omega)} > 3^m$, the symbolic trajectory is linearly contractive.

This motivates the Beatty threshold sequence

$$A020914(m) = \lfloor m \log_2 3 \rfloor + 1,$$

which is the minimal integer k satisfying

$$2^k > 3^m.$$

Consequently, $A020914(m)$ represents the minimal total division count for which the affine slope becomes strictly contractive.

3.2 Example

Consider convergent segments with $m = 4$ iterates of f. The minimal total division count for $m = 4$ is

$$A020914(4) = 7.$$

One realizable division word attaining this minimum is

$$\omega = (1, 1, 1, 4),$$

for which

$$D(\omega) = 1 + 1 + 1 + 4 = 7.$$

Composing the accelerated map according to ω yields

$$f_\omega(x) = \frac{3^4 x + c(\omega)}{2^7} = \frac{81x + 65}{128},$$

so the canonical affine constant for this word is

$$c(\omega) = 65.$$

The integrality condition

$$81x + 65 \equiv 0 \pmod{128}$$

determines a unique residue class

$$x \equiv 15 \pmod{128}.$$

This congruence defines a constraint fiber with modulus 2^7. These congruence classes will be interpreted as symbolic cylinders associated with the division word ω.

Interpretation. The word $\omega = (1, 1, 1, 4)$ means:

- At each of the first three iterates of f, $3x + 1$ contains exactly one factor of 2.
- At the fourth iterate of f, $3x + 1$ contains four factors of 2.

Under the accelerated map

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

Thus $d_3 = 4$ means that the fourth iterate divides by 2^4 in that step. The constant $c(\omega)$ is fixed entirely by this symbolic structure and ensures that precisely the integers in the residue class $x \equiv 15 \pmod{128}$ realize the affine transformation associated with ω.

This example illustrates that the behavior of trajectories is governed entirely by the symbolic data encoded in ω and its associated affine map. The initial values that realize a given word are precisely those satisfying the induced congruence constraint, independent of any particular choice of representative.

The affine representation converts symbolic trajectory data into explicit arithmetic constraints modulo powers of two, allowing realizable trajectories to be studied through congruence geometry rather than direct iteration.

In general, not every division word arises from an actual trajectory; the condition for realizability will be formulated later as an explicit congruence constraint, defining the class of *admissible* words. The next section establishes that admissible words determine unique residue classes.

4 Affine Symbolic Admissibility Grammar

Remark (Division Words as a Dynamical Encoding). Recall from Section 2 that a *division word* $\omega = (d_0, \ldots, d_{m-1})$ records the exact 2-adic valuation $d_i = v_2(3x_i + 1)$, occurring at each iterate of the fully accelerated Collatz map.

Unlike a purely combinatorial encoding, division words retain essential dynamical information. Each division word $\omega = (d_0, \ldots, d_{m-1})$ determines the sequence of multiplicative factors of the linear coefficient

$$r_i = \frac{3}{2^{d_i}},$$

so that the multiplicative growth or decay contribution of each iterate is preserved in the linear coefficient (with the affine term handled separately via $c(\omega)$). This structure enables quantitative control of symbolic trajectories via affine growth and later permits the formulation of arithmetic realizability and descent conditions, even though individual trajectories are collapsed under the encoding.

4.1 Explicit Form of the Affine Constant

Lemma 4.1 (Explicit Form of the Affine Constant)**.** *Let $\omega = (d_0, \ldots, d_{m-1})$ be a division word. Then the affine map*

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

has the explicit form

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i},$$

where empty sums are defined as 0*.*

Proof. Iterating the accelerated map symbolically gives

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}.$$

Repeated substitution yields

$$x_m = \frac{3^m x_0}{2^{D(\omega)}} + \sum_{j=0}^{m-1} \frac{3^{m-1-j}}{2^{\sum_{i=j}^{m-1} d_i}}.$$

Multiplying by $2^{D(\omega)}$ gives

$$2^{D(\omega)}x_m = 3^m x_0 + \sum_{j=0}^{m-1} 3^{m-1-j} 2^{\sum_{i=0}^{j-1} d_i},$$

which identifies the affine constant $c(\omega)$. $\square$

Remark. The affine constant $c(\omega)$ records the cumulative contribution of all additive terms generated during symbolic iteration. Earlier additive contributions are propagated through more subsequent multiplications by 3, while later contributions accumulate more of the preceding dyadic scaling.

Explicit computational examples and a direct reconstruction of the associated affine congruence structure are provided in the appendices.

5 Stepwise Realizability and Admissibility

5.1 Stepwise Realizability

Theorem 5.1 (Affine Stepwise Identity)**.** *Let $\omega = (d_0, \ldots, d_{m-1})$ be a division word, and define $c(\omega)$ by*

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i}.$$

Then for any sequence $(x_0, \ldots, x_m)$ satisfying

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}, \quad 0 \le i < m,$$

the identity

$$2^{D(\omega)}x_m = 3^m x_0 + c(\omega)$$

holds.

Corollary 5.2 (Stepwise Integrality Criterion)**.** *The sequence $(x_0, \ldots, x_m)$ has all intermediate values integral under the accelerated Collatz relations if and only if*

$$3^m x_0 + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Remark (Structural Interpretation). Each term in $c(\omega)$ records the propagated contribution of a single additive "+1" created during iteration. Earlier contributions accumulate additional powers of 3 through forward propagation and additional powers of 2 through delayed subsequent divisions by powers of two, producing a fully time-ordered encoding of the trajectory in a single affine constant.

5.2 Division Word Valuation

Lemma 5.3 (Exact Valuation Congruence). *Let x be an odd integer and let $d \geq 1$. Then*

$$v_2(3x+1) = d$$

if and only if

$$3x+1 \equiv 2^d \pmod{2^{d+1}}.$$

Equivalently,

$$3x+1 = 2^d(2k+1)$$

for some integer k.

Proof. Suppose first that

$$v_2(3x+1) = d.$$

Then by definition,

$$3x+1 = 2^d y,$$

where y is odd. Writing

$$y = 2k+1,$$

gives

$$3x+1 = 2^d(2k+1) = 2^{d+1}k + 2^d,$$

and therefore

$$3x+1 \equiv 2^d \pmod{2^{d+1}}.$$

Conversely, suppose that

$$3x+1 \equiv 2^d \pmod{2^{d+1}}.$$

Then there exists an integer k such that

$$3x+1 = 2^{d+1}k + 2^d = 2^d(2k+1).$$

Since $2k+1$ is odd, exactly d factors of 2 divide $3x+1$. Hence

$$v_2(3x+1) = d.$$

$\square$

Corollary 5.4 (Recursive Exact-Valuation Refinement). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be an admissible division word and let

$$\omega_j = (d_0, \dots, d_{j-1})$$

denote its prefixes.

For each j, the condition

$$v_2(3f^j(x)+1) = d_j$$

is equivalent to the congruence

$$3f^j(x) + 1 \equiv 2^{d_j} \pmod{2^{d_j+1}}.$$

Consequently, each additional valuation constraint refines the realization set of the prefix ω_j to a smaller realization set associated with the extended prefix

$$\omega_{j+1}.$$

Thus admissible division words generate a nested sequence of arithmetic constraints, and extensions of division words correspond to successive refinements of residue classes.

Remark. The corollary formalizes the local nature of admissibility. Exact valuation constraints are imposed one step at a time and depend only on the current accelerated iterate. Consequently, realizability propagates forward recursively through the valuation sequence rather than being determined solely by a terminal congruence condition.

Theorem 5.5 (Unique Residue Lifting). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be an admissible division word. Then there exists a unique residue class

$$r_\omega \pmod{2^{D(\omega)}}$$

consisting precisely of the integers realizing ω*. Moreover, if*

$$\omega_j = (d_0, \dots, d_{j-1})$$

denotes the prefix of length j*, then the corresponding residue classes satisfy*

$$r_{\omega_{j+1}} \equiv r_{\omega_j} \pmod{2^{D(\omega_j)}}.$$

Thus each additional valuation constraint uniquely refines the residue class of its prefix.

Proof. We proceed by induction on the length m of the division word. For $m = 1$, the word consists of a single valuation

$$\omega = (d_0).$$

By Lemma 5.3,

$$v_2(3x + 1) = d_0$$

if and only if

$$3x + 1 \equiv 2^{d_0} \pmod{2^{d_0+1}}.$$

Since 3 is invertible modulo every power of two, this congruence has a unique solution modulo 2^{d_0+1}. Thus the realization set of ω is a single residue class.

Assume now that the result holds for all admissible words of length m, and let

$$\omega' = (d_0, \dots, d_m)$$

be an admissible word of length $m + 1$. Let

$$\omega = (d_0, \dots, d_{m-1})$$

be its length-m prefix. By the induction hypothesis, the integers realizing ω form a unique residue class

$$r_\omega \pmod{2^{D(\omega)}}.$$

The additional valuation condition

$$v_2(3f^m(x) + 1) = d_m$$

is equivalent, by Lemma 5.3, to

$$3f^m(x) + 1 \equiv 2^{d_m} \pmod{2^{d_m+1}}.$$

Using the affine representation of the prefix,

$$f^m(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

this becomes a linear congruence condition on x modulo

$$2^{D(\omega)+d_m} = 2^{D(\omega')}.$$

Since the coefficient of x remains 3^m, which is odd and therefore invertible modulo powers of two, this congruence determines a unique refinement of the residue class

$$r_\omega \pmod{2^{D(\omega)}}$$

to a residue class

$$r_{\omega'} \pmod{2^{D(\omega')}}.$$

Therefore every admissible extension uniquely refines the residue class of its prefix. The result follows by induction. □

5.3 Affine Realization

Theorem 5.6 (Affine Realization Theorem). *Let ω be a division word.*

An integer x realizes ω if and only if

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Proof. Assume that

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

By the Exact Valuation Congruence Lemma 5.3, the residue class associated with the full word ω is contained in the residue class of every prefix

$$\omega_j = (d_0, \dots, d_{j-1}).$$

Therefore, for every prefix length j,

$$x \equiv r_{\omega_j} \pmod{2^{D(\omega_j)}}.$$

We prove inductively that the successive iterates generated from x realize the prescribed valuations. For $j = 0$, the prefix condition gives

$$x \equiv r_{(d_0)} \pmod{2^{d_0}},$$

and therefore

$$2^{d_0} \mid 3x + 1.$$

Because the residue class is defined using the exact division count d_0, the next higher power does not divide:

$$2^{d_0+1} \nmid 3x + 1.$$

Hence

$$v_2(3x + 1) = d_0.$$

Therefore the first accelerated iterate is

$$x_1 = \frac{3x_0 + 1}{2^{d_0}},$$

and is an integer. Now assume that the first j steps have been realized, so that

$$x_j = f^j(x)$$

is the integer determined by the prefix

$$(d_0, \ldots, d_{j-1}).$$

The prefix congruence condition for

$$\omega_{j+1} = (d_0, \ldots, d_j)$$

implies that

$$2^{d_j} \mid 3x_j + 1.$$

Again, exactness of the prefix residue condition excludes the higher power:

$$2^{d_j+1} \nmid 3x_j + 1.$$

Consequently,

$$v_2(3x_j + 1) = d_j,$$

and the next iterate

$$x_{j+1} = \frac{3x_j + 1}{2^{d_j}}$$

is an integer. By induction, every step satisfies the prescribed valuation constraint, and therefore

$$v_2(3f^j(x) + 1) = d_j, \qquad 0 \le j < m.$$

Hence x realizes the division word ω. $\square$

5.4 Admissibility of Division Words

Definition 5.1 (Admissible Division Word)**.** A division word

$$\omega = (d_0, \ldots, d_{m-1})$$

is admissible if its induced affine residue class contains an integer whose successive iterates satisfy

$$v_2(3x_i + 1) = d_i, \qquad 0 \le i < m.$$

Remark. Admissibility is the condition that the formal division word corresponds to an actual finite valuation trajectory of the accelerated Collatz map.

5.5 Residue Class Representation of Division Words

Theorem 5.7 (Affine Constraint Representation)**.** *Let ω be an admissible division word of length m.*

Then there exists a unique residue $r_\omega \pmod{2^{D(\omega)}}$ such that

$$x \text{ satisfies } 3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}} \iff x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Proof. From the affine representation,

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

integrality is equivalent to

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is odd, it is invertible modulo $2^{D(\omega)}$, yielding

$$x \equiv -c(\omega) \cdot (3^m)^{-1} \pmod{2^{D(\omega)}}.$$

This determines a unique residue modulo $2^{D(\omega)}$.

Conversely, any x in this class satisfies the integrality condition. By Theorem 5.1, this is equivalent to the existence of a trajectory whose successive iterates satisfy

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}, \quad 0 \le i < m,$$

and hence realize the division word ω.

Thus admissible division words are in one-to-one correspondence with affine congruence constraints modulo powers of two. $\square$

Corollary 5.8. *An integer x realizes ω if and only if*

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Lemma 5.9 (Prefix Nesting of Realization Sets)**.** *Let ω, ω' be admissible division words.*

If the realization sets of ω and ω' intersect, then one word is a prefix of the other.

Proof. Every integer x determines a unique infinite valuation sequence

$$(v_2(3x+1),\, v_2(3f(x)+1),\, v_2(3f^2(x)+1), \dots).$$

If x lies in the realization sets of both ω and ω', then both words must agree with the initial segment of this valuation sequence.

Therefore the two words cannot diverge at any position before the shorter word terminates. Hence one word is a prefix of the other. $\square$

6 Cylinder Geometry of the Affine Grammar

The admissible division-word grammar is organized by 2-adic residue classes induced by admissible division words.

These residue classes provide an arithmetic realization of the symbolic dynamics and naturally suggest a geometric interpretation in terms of nested congruence classes, developed below.

6.1 Cylinder of a Division Word

Definition 6.1 (Cylinder)**.** Let ω be an admissible division word with associated residue class

$$r_\omega \pmod{2^{D(\omega)}}.$$

The cylinder associated with ω is

$$C_\omega := \{x \in \mathbb{N} : x \equiv r_\omega \pmod{2^{D(\omega)}}\}.$$

6.2 Prefix Structure and Forest Geometry of Cylinders

Lemma 6.1 (Prefix Nesting of Cylinders)**.** *Let ω and ω' be admissible division words.*

If

$$C_\omega \cap C_{\omega'} \neq \varnothing,$$

then one of the words is a prefix of the other.

Proof. Suppose

$$x \in C_\omega \cap C_{\omega'}.$$

Then x realizes both division words. By the Affine Realization Theorem, every integer determines a unique infinite valuation sequence

$$(v_2(3x+1),\, v_2(3f(x)+1),\, v_2(3f^2(x)+1), \dots).$$

Since both ω and ω' are finite prefixes of this same valuation sequence, they cannot differ before the shorter word terminates. Hence one word is a prefix of the other. □

Theorem 6.2 (Cylinder Inclusion Criterion)**.** *Let ω and η be admissible division words.*

Then

$$C_\eta \subseteq C_\omega$$

if and only if

$$\omega$$

is a prefix of

$$\eta.$$

Proof. Suppose first that ω is a prefix of η. Any integer realizing η must realize every prefix of η, including ω. Therefore

$$C_\eta \subseteq C_\omega.$$

Conversely, suppose

$$C_\eta \subseteq C_\omega.$$

Choose

$$x \in C_\eta.$$

Then

$$x \in C_\eta \cap C_\omega.$$

By Lemma 6.1, one word is a prefix of the other. If η were a prefix of ω, then applying the forward direction already established would imply

$$C_\omega \subseteq C_\eta.$$

Together with

$$C_\eta \subseteq C_\omega,$$

this would force

$$C_\eta = C_\omega,$$

Since equal cylinders have equal moduli, we obtain

$$D(\eta) = D(\omega).$$

Because one word is a prefix of the other and the two words have equal depth, they must coincide:

$$\eta = \omega.$$

Therefore ω must be a prefix of η. $\square$

Corollary 6.3 (Disjointness of Sibling Cylinders)**.** *If neither admissible word is a prefix of the other, then*

$$C_\omega \cap C_\eta = \varnothing.$$

Proof. If the cylinders intersected, then by Lemma 6.1, one word would be a prefix of the other, contradicting the assumption. $\square$

Corollary 6.4 (Disjointness at Equal Depth)**.** *Distinct admissible division words of equal length define disjoint cylinders.*

Proof. If two equal-length words intersected, then by Lemma 6.1, one would be a prefix of the other. Equal lengths force equality of the words, contradicting distinctness. □

Corollary 6.5 (Symbolic-Arithmetic Correspondence)**.**

$$\omega \preceq \eta \iff C_\eta \subseteq C_\omega.$$

Proof. This is exactly the content of Theorem 6.2. □

Remark. The previous results identify symbolic ancestry with arithmetic containment. Thus admissible division words organize into a collection of nested 2-adic cylinders whose inclusion relations coincide exactly with the prefix order on division words.

The admissible symbolic grammar therefore forms a disjoint forest of prefix trees rather than a single global tree. Each node corresponds to an admissible one-step refinement of a cylinder, obtained by extending the associated division word by one admissible symbol. Each descendant cylinder is obtained by refining the associated congruence class modulo a higher power of two.

6.3 Contractive Cylinders

Definition 6.2 (Contractive Cylinder)**.** An admissible cylinder C_ω is called contractive if the associated affine map

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}$$

has contractive linear coefficient:

$$\frac{3^{|\omega|}}{2^{D(\omega)}} < 1.$$

Equivalently,

$$D(\omega) > |\omega| \log_2 3.$$

6.4 Cylinder Contraction Threshold

Lemma 6.6 (Minimal Contraction Threshold)**.** *Let ω be an admissible division word of length m. If the associated affine map*

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

is contractive in its linear coefficient, then

$$D(\omega) \geq A020914(m),$$

where

$$A020914(m) = \lfloor m \log_2 3 \rfloor + 1$$

is the minimal integer k satisfying

$$2^k > 3^m.$$

Proof. Contractivity requires

$$\frac{3^m}{2^{D(\omega)}} < 1.$$

which is equivalent to

$$2^{D(\omega)} > 3^m.$$

The minimal integer satisfying this inequality is exactly $A020914(m)$, so

$$D(\omega) \geq A020914(m).$$

□

Remark. The Beatty threshold sequence A020914 governs the growth/decay balance along admissible branches within every cylinder of the forest.

7 Beatty Refinement of the Cylinder Forest

The prefix-tree structure of admissible cylinders induces a corresponding symbolic grammar on division words. The admissible extensions of a word are governed by cumulative dyadic growth constraints, determined by the minimal contraction threshold sequence A020914.

The branching structure of minimally contractive division words is governed by the prefix-sum constraints induced by

$$B(n) = A020914(n) = \lfloor n \log_2 3 \rfloor + 1.$$

This sequence plays a dual role: it is simultaneously the global Beatty threshold for dyadic contraction and the local admissibility boundary governing extension of division words.

Remark (Beatty-Controlled Refinement). Every admissible division word determines its admissible one-symbol extensions. Since admissibility is governed by the Beatty thresholds $B(m)$, the entire cylinder forest is generated recursively from the Beatty increment sequence.

7.1 Propagation Versus Expansion

The increment sequence

$$\delta(n) = B(n+1) - B(n)$$

takes values in $\{1, 2\}$.

The distinction between these two increment types governs the combinatorial evolution of the admissible grammar. In cylinder language, these increments correspond to whether refinement occurs within an existing residue class or splits it into new congruence classes

Lemma 7.1 (Unit Increments Preserve the Existing Grammar)**.** *Suppose*

$$B(n+1) - B(n) = 1.$$

introduces no new interior admissibility directions beyond propagation of existing prefixes. The admissible words at level $n + 1$ *arise solely from propagation of previously admissible prefix configurations.*

Structural mechanism. A unit increment increases the terminal admissibility bound by exactly one:

$$B(n+1) = B(n) + 1.$$

Since every division word component satisfies $d_i \geq 1$, the final coordinate is forced to absorb precisely this additional unit.

No new slack is introduced into the earlier prefix constraints

$$H(i) < B(i+1),$$

since the admissibility threshold increases by exactly one, the mandatory contribution of the new symbol exhausts the additional allowance. Consequently every admissible extension is obtained simply by extending an existing admissible prefix. No additional extension pattern becomes feasible.

Thus the grammar propagates existing admissible configurations and every admissible extension is obtained by prolonging an existing admissible prefix; no additional admissible branch is created. □

Lemma 7.2 (Double Increments Expand the Existing Grammar)**.** *Suppose*

$$B(n+1) - B(n) = 2.$$

introduces incremental slack into the terminal admissibility bound, providing additional internal directions for admissible extensions. New admissible words at level $n+1$ *arise from both propagation of previously admissible prefix configurations and from the new admissible directions created by the additional slack.*

Structural mechanism. A double increment increases the terminal admissibility bound by exactly two:

$$B(n+1) = B(n) + 2.$$

Since every division word component satisfies $d_i \geq 1$, the final coordinate absorbs precisely one unit associated with the existing admissible division words, leaving a remaining slack of one.

Only one unit is absorbed into the earlier prefix constraints

$$H(i) < B(i+1),$$

which leaves one additional unit beyond the mandatory contribution of the terminal symbol. This additional freedom permits extension patterns that were previously excluded by the Beatty bound.

Thus the grammar propagates existing admissible configurations by prolonging an existing admissible prefix; while the incremental unit of slack means a new admissible branch is created. □

Theorem 7.3 (Grammar Expansion)**.** *The admissible cylinder forest is generated recursively from the Beatty threshold sequence. At each level, unit Beatty increments introduce only propagated extensions of existing admissible words, whereas double Beatty increments permit additional admissible extensions beyond simple propagation.*

8 Symbolic Growth and Spectral Complexity

The cylinder forest introduced in the previous section possesses two competing forms of complexity:

1. combinatorial growth in the number of admissible cylinders, and
2. dyadic contraction in the size of individual cylinders.

The first of these is purely symbolic and is quantified below.

8.1 Symbolic complexity

Definition 8.1 (Symbolic complexity)**.** Let

$$N(m)$$

denote the number of admissible division words of length m. Equivalently, $N(m)$ is the number of admissible cylinders appearing at depth m in the symbolic forest.

Remark. The quantity $N(m)$ measures the combinatorial complexity of the admissible grammar at depth m. As the forest evolves under symbolic refinement, $N(m)$ records the number of distinct admissible branches that survive to depth m.

8.2 Spectral entropy

Definition 8.2 (Symbolic growth rate)**.** Define the symbolic growth rate of the admissible grammar by

$$\lambda := \limsup_{m\to\infty} N(m)^{1/m}.$$

Remark. The quantity λ is the exponential branching rate of the admissible cylinder forest. It measures the asymptotic combinatorial complexity of admissible symbolic extensions.

Lemma 8.1 (Uniform exponential envelope). *For every $\varepsilon > 0$, there exists m_0 such that*

$$N(m) \le (\lambda + \varepsilon)^m$$

for all $m \ge m_0$.

Proof. This is an immediate consequence of the definition of

$$\lambda := \limsup_{m\to\infty} N(m)^{1/m}.$$

$\square$

8.3 Entropy versus dyadic contraction

Each admissible cylinder C_ω has natural density

$$\mu(C_\omega) = 2^{-D(\omega)}.$$

From Lemma 6.6

$$D(\omega) \ge B(m)$$

where

$$B(m) = A020914(m) = \lfloor m \log_2 3 \rfloor + 1$$

therefore

$$2^{-D(\omega)} \le 2^{-B(m)} < 3^{-m}.$$

Consequently, the total measure carried by the depth-m residual cylinders is bounded above by the product of two competing factors:

- the number $N(m)$ of admissible cylinders at depth m, and
- the maximal measure of an admissible cylinder at depth m.

The following section combines these two ingredients to estimate the residual mass of the cylinder forest. Since cylinders at equal depth are pairwise disjoint (Corollary 6.4), their measures add. Consequently, estimating the residual mass reduces to estimating the product of the number of admissible cylinders and the maximal measure of any one cylinder.

9 Measure Collapse of the Residual Forest

The admissible forest constructed in the previous sections naturally induces a nested family of measurable subsets of the odd integers. These sets encode the integers whose symbolic trajectories remain admissible to increasing depths.

The central question is whether the densities

$$\mu(S_m)$$

remain bounded away from zero as

$$m \to \infty,$$

or whether repeated refinement causes the residual forest to collapse to a set of zero density.

9.1 Residual cylinder sets

Definition 9.1 (Depth-m residual set)**.** Let

$$S_m := \bigcup_{|\omega|=m} C_\omega$$

where the union ranges over all admissible division words of length m.

Remark. The set S_m consists of integers whose valuation trajectories admit an admissible division-word prefix of length m. Equivalently, these are precisely the trajectories that remain unresolved after m refinement steps.

9.2 First-passage layers

Definition 9.2 (First-passage layer)**.** Define the first-passage layer at depth m by

$$F_m := S_m \setminus S_{m+1}.$$

Remark. The set F_m consists of integers whose symbolic trajectories admit an admissible prefix of length m, but admit no admissible extension to depth $m+1$.

Lemma 9.1 (Disjoint residual decomposition)**.** *For every $m \geq 1$,*

$$S_m = F_m \sqcup S_{m+1},$$

where $\sqcup$ denotes disjoint union.

Proof. By definition,

$$F_m = S_m \setminus S_{m+1}.$$

Hence every element of S_m either:

- remains unresolved to depth $m+1$, or
- exits the residual forest through the first-passage layer F_m

These two possibilities are mutually exclusive, giving the disjoint decomposition. $\square$

Remark (Conservation of symbolic mass). The decomposition

$$S_m = F_m \sqcup S_{m+1}$$

shows that symbolic refinement conserves total mass: cylinders are not destroyed under refinement, but redistributed between deeper residual cylinders and first-passage escape cylinders.

Corollary 9.2 (Global first-passage decomposition). *The initial residual set admits the disjoint decomposition*

$$S_1 = \left(\bigsqcup_{m \geq 1} F_m \right) \sqcup S_\infty,$$

where

$$S_\infty = \bigcap_{m \geq 1} S_m$$

is the infinite residual set.

Proof. Iterating the decomposition

$$S_m = F_m \sqcup S_{m+1}$$

gives

$$S_1 = F_1 \sqcup F_2 \sqcup \cdots \sqcup F_m \sqcup S_{m+1}.$$

Passing to the limit $m \to \infty$ yields

$$S_1 = \left(\bigsqcup_{m \geq 1} F_m \right) \sqcup S_\infty. \qquad \square$$

Remark (Symbolic conservation law)*.* The global decomposition shows that every residual trajectory either:

- eventually exits into a first-passage layer, or
- survives indefinitely in S_∞.

No symbolic mass is lost or duplicated under refinement; it is partitioned disjointly between finite first-passage escape layers and the infinite residual set.

9.3 Cylinder measure

Definition 9.3 (Natural dyadic measure)**.** Let C_ω be a cylinder determined by a residue class modulo $2^{D(\omega)}$. Define its measure by

$$\mu(C_\omega) = 2^{-D(\omega)}.$$

Remark. Since each admissible cylinder corresponds to a single residue class modulo $2^{D(\omega)}$, its canonical dyadic weight is proportional to $2^{-D(\omega)}$. Up to normalization of the odd integers, this coincides with the dyadic scaling above.

Lemma 9.3 (Disjointness at fixed depth)**.** *Distinct admissible cylinders of equal depth are disjoint.*

Proof. This follows from the prefix-nesting property established in Section 6. □

Corollary 9.4 (Measure decomposition)**.** *For every m,*

$$\mu(S_m) = \sum_{|\omega|=m} 2^{-D(\omega)}.$$

Remark (Conservation under symbolic refinement)*.* The first-passage decomposition partitions the initial residual set into disjoint escape layers together with the infinite residual set. Thus dyadic mass is neither created nor lost under refinement: it is redistributed between progressively finer symbolic layers.

9.4 Nested residual structure

Definition 9.4 (Infinite residual set)**.** Define the infinite residual set by

$$S_\infty := \bigcap_{m=1}^{\infty} S_m.$$

Remark. The set S_∞ consists of integers whose symbolic trajectories admit admissible extensions at every finite depth, equivalently, integers which never exit the residual forest under successive refinement.

Theorem 9.5 (Continuity of measure under nested refinement)**.** *Since*

$$S_{m+1} \subseteq S_m,$$

the measures satisfy

$$\mu(S_\infty) = \lim_{m\to\infty} \mu(S_m).$$

Proof. This is the standard continuity property of measure for nested decreasing measurable sets. ☐

9.5 Residual Mass Collapse

Theorem 9.6 (Residual Mass Estimate)**.** *For every depth m,*

$$\mu(S_m) = \sum_{|\omega|=m} 2^{-D(\omega)} \le N(m)\, 2^{-B(m)},$$

where

$$B(m) = A020914(m) = \lfloor m \log_2 3 \rfloor + 1.$$

Proof. By Corollary 9.4,

$$\mu(S_m) = \sum_{|\omega|=m} 2^{-D(\omega)}.$$

By Lemma 6.6,

$$D(\omega) \ge B(m)$$

for every admissible division word of length m. Therefore

$$2^{-D(\omega)} \le 2^{-B(m)}.$$

Since the sum ranges over $N(m)$ admissible division words of length m,

$$\mu(S_m) \le N(m)\, 2^{-B(m)},$$

as claimed. ☐

Theorem 9.7 (Residual Mass Collapse)**.** *Suppose the symbolic growth rate satisfies*

$$\lambda < 3,$$

where

$$\lambda = \limsup_{m\to\infty} N(m)^{1/m}.$$

Then

$$\mu(S_m) \longrightarrow 0.$$

Consequently,

$$\mu(S_\infty) = 0.$$

Proof. Let

$$\varepsilon > 0$$

be chosen so that

$$\lambda + \varepsilon < 3.$$

By Lemma 8.1, there exists m_0 such that

$$N(m) \le (\lambda + \varepsilon)^m$$

for every $m \ge m_0$. Combining this estimate with Theorem 9.6 gives

$$\mu(S_m) \le (\lambda + \varepsilon)^m \, 2^{-B(m)}.$$

Since

$$2^{B(m)} > 3^m,$$

we have

$$2^{-B(m)} < 3^{-m}.$$

Hence

$$\mu(S_m) < \left(\frac{\lambda + \varepsilon}{3}\right)^m.$$

Because

$$\frac{\lambda + \varepsilon}{3} < 1,$$

the right-hand side converges exponentially to zero. Therefore

$$\mu(S_m) \to 0.$$

Finally, Theorem 9.5 implies

$$\mu(S_\infty) = \lim_{m\to\infty} \mu(S_m) = 0.$$

□

9.6 First-Passage Completeness

Theorem 9.8 (First-Passage Completeness)**.** *Suppose*

$$\lambda < 3.$$

Then

$$\mu(S_1) = \sum_{m=1}^{\infty} \mu(F_m).$$

In particular, if the initial residual set is normalized so that

$$\mu(S_1) = 1,$$

then

$$\sum_{m=1}^{\infty} \mu(F_m) = 1.$$

Thus the set of integers which remain in the residual forest indefinitely has natural density zero.

Proof. By Corollary 9.2,

$$S_1 = \left(\bigsqcup_{m \geq 1} F_m \right) \sqcup S_\infty.$$

Since the union is disjoint,

$$\mu(S_1) = \sum_{m \geq 1} \mu(F_m) + \mu(S_\infty).$$

By Theorem 9.7,

$$\mu(S_\infty) = 0.$$

Therefore

$$\mu(S_1) = \sum_{m \geq 1} \mu(F_m).$$

If the measure is normalized so that $\mu(S_1) = 1$, the stated identity follows immediately. $\square$

9.7 Interpretation

The preceding results reduce the density-one problem for accelerated Collatz trajectories to a purely symbolic question. Every measure-theoretic aspect of the argument is now contained in Theorem 9.7. Consequently, it remains only to establish that the symbolic growth rate of the admissible grammar satisfies

$$\lambda < 3$$

The remainder of the paper is devoted entirely to proving this spectral inequality.

10 Extremal Growth in the Sturmian Return-Word System

10.1 Dyadic Threshold Sequence Sturmian Structure

Theorem 10.1 (Sturmian realization of the threshold grammar)**.** *Let*

$$B(n) = \lfloor n \log_2 3 \rfloor + 1.$$

Then the increment sequence

$$\delta(n) = B(n+1) - B(n)$$

is a Sturmian word over the alphabet $\{1, 2\}$*. Consequently the admissible grammar inherits the combinatorial structure of an irrational rotation.*

Proof. This is the classical characterization of Beatty increment sequences associated with irrational slopes. □

Let

$$\alpha := \log_2 3, \qquad d(n) := \lfloor n\alpha \rfloor + 1.$$

The increment sequence

$$\delta(n) := d(n+1) - d(n) \in \{1, 2\}$$

is a Sturmian word of slope $\alpha - 1$ and intercept 0. Equivalently, δ defines a symbolic dynamical system

$$X_\alpha \subset \{1, 2\}^{\mathbb{N}}$$

generated by an irrational rotation. All admissible growth structures considered in this paper are functions of this Sturmian coding.

10.2 Return Words and Primitive Cycle Decomposition

Definition 10.1 (Return Words)**.** A finite word C is a return word of the Sturmian system X_α if it is the minimal block between successive visits to a fixed cylinder set induced by δ. We denote by $\mathcal{R}_\alpha$ the set of return words.

Theorem 10.2 (Return Words via Continued Fractions)**.** *Let $\alpha = \log_2 3$ and let $\delta(n)$ be the Sturmian coding induced by rotation R_α.*

Let q_k denote the denominators of the continued-fraction convergents of α. For the prefix defining the extremal admissible growth process, the set of return words is exactly:

$$\mathcal{R}_\alpha = \{C_{q_k}, C_{q_{k+1}}\}$$

for some index k determined by the chosen prefix. In particular, for the admissible threshold structure of $d(n) = \lfloor n\alpha \rfloor + 1$, the relevant convergent pair is:

$$(q_k, q_{k+1}) = (5, 12).$$

Proof. The sequence $\delta(n)$ is a Sturmian word of slope α. Return words to any prefix correspond to first-return maps of the rotation R_α to a subinterval I.

The convergent denominators begin (see Appendix B for details):

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\ 31867,\ 79335, \ldots$$

For the prefix relevant to the admissible grammar, the associated return times are the consecutive denominators 5 and 12. Hence the return-word set is $\{C_5, C_{12}\}$. □

Although the shortest return word is the single double increment $C_1 = (2)$, repeated doubles generate the Catalan sequence $1, 2, 5, 14, 42, 132, \ldots$ whose asymptotic exponential growth rate is 4. Since this exceeds the critical threshold 3, the pure-double process cannot serve as a useful comparison envelope.

Proposition 10.1 (Extremal Primitive Comparison Word)**.** The return word C_5 is the shortest return word whose induced comparison growth is strictly less than 3.

Remark (Diophantine Origin of the Grammar)*.* The admissible grammar is governed not by arbitrary combinatorial rules but by the Diophantine approximation properties of $\log_2 3$. The convergent denominators

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\ 31867,\ 79335, \ldots$$

arise from the continued-fraction approximants of

$$\alpha = \log_2 3,$$

and consequently, the primitive return words governing admissible refinement are determined by best rational approximations to the Beatty slope rather than by ad hoc combinatorial choices.

$$\frac{p_k}{q_k} \approx \log_2 3.$$

Equivalently,

$$2^{p_k} \approx 3^{q_k},$$

so the associated multiplicative cocycle satisfies

$$\frac{2^{p_k}}{3^{q_k}} \to 1.$$

In particular, the appearance of the 5-, 12-, 41-, and 53-cycles is not accidental; these cycles are induced by the Diophantine approximation hierarchy of $\log_2 3$.

Lemma 10.3 (Induced Return-Word Hierarchy)**.** *Every admissible word in the Sturmian language generated by*

$$\delta(n) = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$$

admits a decomposition into return words generated recursively from the primitive pair (C_5, C_{12})*.*

10.3 Multiplicative Cocycle Structure

Define the multiplicative growth cocycle

$$G(w) := \frac{2^{\#s(w)+2\#d(w)}}{3^{|w|}},$$

where $\#s(w)$ and $\#d(w)$ are the number of single and double increments in w.

Lemma 10.4 (Cocycle Additivity)**.** *For concatenation of admissible return words,*

$$G(C_{i_1} \cdots C_{i_n}) = \prod_{k=1}^{n} G(C_{i_k}).$$

Equivalently,

$$\log G(w) = \sum_k \log G(C_{i_k}).$$

Thus the global exponential growth rate reduces to weighted averages of return-word contributions.

10.4 Local Evaluation of Return Words

The local comparison constants computed in Appendix C yield the following per-symbol cocycle contributions. The two primitive return words satisfy:

$$\frac{1}{5}\log G(C_5) = \log\left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5},$$

$$\frac{1}{12}\log G(C_{12}) < \frac{1}{5}\log G(C_5).$$

Hence C_5 is the unique extremal return word with maximal per-symbol drift.

Proposition 10.2 (Extremal Cocycle Principle)**.** For any admissible infinite word $\delta \in X_\alpha$,

$$\limsup_{n\to\infty} \frac{1}{n}\log G(\delta_{[1,n]}) \leq \frac{1}{5}\log G(C_5).$$

Proof. By the unique decomposition into return words and cocycle additivity, the growth rate is a convex combination of the per-symbol contributions of C_5 and C_{12}.

Since

$$\frac{1}{5}\log G(C_5) > \frac{1}{12}\log G(C_{12}),$$

the maximal comparison average is obtained by replacing every return word with C_5, corresponding formally to the periodic comparison word C_5^∞. □

10.5 Global Growth Bound

Theorem 10.5 (Spectral Growth Bound)**.** *Let λ denote the symbolic growth rate introduced in Section 8.*

Then

$$\lambda \leq \exp\left(\frac{1}{5}\log G(C_5)\right) < 3.$$

Proof. By Proposition 10.1, every admissible infinite word has average cocycle growth bounded above by that of C_5^∞. Thus

$$\lambda \le \exp\left(\frac{1}{5}\log G(C_5)\right).$$

Via explicit evaluation of the local cocycle contributions (Appendix C),

$$\left(3^2 \cdot \frac{123}{33} \cdot \frac{341}{131} \cdot \frac{329}{123}\right)^{1/5} < 3.$$

□

Theorem 10.5 closes the logical chain developed throughout the paper. The Beatty threshold sequence induces the admissible grammar; the grammar determines the symbolic growth rate λ; the Sturmian return-word analysis establishes the strict bound $\lambda < 3$; and Theorem 9.7 then forces the residual forest to have zero natural density. Thus the symbolic, arithmetic, and measure-theoretic components of the argument fit together in a single chain of implications.

11 Conclusion

This paper reformulates the accelerated Collatz problem in terms of an admissible symbolic grammar whose arithmetic realization is given by nested dyadic cylinders. The resulting cylinder forest provides an exact symbolic partition of the odd integers together with a natural measure-theoretic filtration.

Within this framework the measure of the residual forest is controlled by two competing exponential processes: symbolic proliferation and dyadic contraction. The measure-theoretic analysis shows that these combine through a single spectral parameter λ. Consequently, proving

$$\lambda < 3$$

is sufficient to force the residual forest to have natural density zero.

Combining the spectral estimate with the residual mass estimate shows that the infinite residual set has natural density zero. Equivalently, the first-passage layers exhaust the entire normalized dyadic measure.

The principal contribution is therefore not merely a new estimate, but a reformulation of the density problem itself. Rather than analyzing individual Collatz trajectories directly, the argument transfers the problem through three equivalent descriptions:

$$\text{symbolic grammar} \longrightarrow \text{cylinder geometry} \longrightarrow \text{measure dynamics.}$$

Within this framework, the density question becomes a problem of controlling the spectral growth of an admissible Sturmian grammar.

This synthesis of symbolic dynamics, Diophantine approximation, and dyadic measure provides a unified framework in which the density behavior of the accelerated Collatz map is governed by a single explicit spectral inequality.

A The Canonical Affine Map

Each division word ω induces an affine map $F_\omega(x)$ of length m (see Section 2)

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}, \qquad D(\omega) = d_0 + \cdots + d_{m-1},$$

where $c(\omega)$ is the affine constant canonically determined by the division word ω. The associated congruence constraint is naturally considered modulo $2^{D(\omega)}$.

Although the existence and uniqueness of $c(\omega)$ follows from the affine representation developed in Section 2, the explicit computation is useful for concrete examples and software implementations. The following procedure reconstructs the same affine constant $c(\omega)$ from any representative of the associated residue class.

A.1 Canonical Choice

For a fixed ω, any integer x in the corresponding residue class satisfies

$$3^m x + c \equiv 0 \pmod{2^{D(\omega)}}.$$

We remove ambiguity by selecting the unique representative

$$0 \le c(\omega) < 2^{D(\omega)}.$$

This choice depends only on ω and is independent of the representative x.

A.2 Step-by-Step Computation

Given ω of length m and total division count $D(\omega)$:

1. Choose any representative x of the residue class determined by ω.
2. Compute $y = 3^m x$.
3. Let $r = \lceil y/2^{D(\omega)} \rceil$ be the smallest integer such that $r \cdot 2^{D(\omega)} \ge y$.
4. Define $c(\omega) = r \cdot 2^{D(\omega)} - y$.

By this construction, $0 \le c(\omega) < 2^{D(\omega)}$ ensuring that

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}} \in \mathbb{Z} \quad \text{for all } x \text{ in the class.}$$

A.3 Example

Let $\omega = (1,1,1,4)$ with $m = 4$ and $D(\omega) = \sum_{i=0}^{3} d_i = 1+1+1+4 = 7$:

1. Choose $x = 15$. Compute $y = 3^4 \cdot 15 = 1215$.
2. $2^7 = 128$, so $r = \lceil 1215/128 \rceil = 10$.
3. $c(\omega) = 10 \cdot 128 - 1215 = 65$.

Hence the canonical map is

$$F_\omega(x) = \frac{81x + 65}{128}.$$

Checking another representative $x = 143$:

$$F_\omega(143) = \frac{81 \cdot 143 + 65}{128} = 91,$$

confirming that the same $c(\omega)$ works for the entire residue class. The admissibility condition becomes

$$81x + 65 \equiv 0 \pmod{128}.$$

Solving yields the unique residue class

$$x \equiv 15 \pmod{128}.$$

Hence the cylinder is

$$x(k) = 15 + 128k.$$

Applying the affine map,

$$F_\omega(15 + 128k) = 81k + 10.$$

Thus the symbolic cylinder generated by ω is mapped onto the arithmetic progression

$$10,\ 91,\ 172,\ 253, \ldots$$

with common difference $81 = 3^4$.

Remark

This construction is purely illustrative and is not required for the density or growth arguments in the main text. It demonstrates concretely how the affine map is fully determined by the division word ω.

By the Cylinder Representation Theorem (Section 6), every realization of ω belongs to a single residue class modulo $2^{D(\omega)}$. Consequently, the value of $c(\omega)$ computed above is independent of the chosen representative.

Remark (Cylinder Rigidity). The division word ω uniquely determines its canonical affine constant $c(\omega)$, and therefore uniquely determines the affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

Since admissible realizations of ω occupy a single residue class modulo $2^{D(\omega)}$, the symbolic data encoded by ω simultaneously determines both the affine dynamics and the associated cylinder.

A.4 Illustrative Derivation of the Affine Constant

For readers wishing to see the affine constant arise directly from iterated substitution, consider a general division word

$$\omega = (a, b, c, d)$$

of length 4.

The accelerated relations are

$$x_1 = \frac{3x_0 + 1}{2^a}, \qquad x_2 = \frac{3x_1 + 1}{2^b}, \qquad x_3 = \frac{3x_2 + 1}{2^c}, \qquad x_4 = \frac{3x_3 + 1}{2^d}.$$

Eliminating intermediate variables recursively yields

$$2^{a+b+c+d} x_4 = 3^4 x_0 + 3^3 + 3^2 2^a + 3\,2^{a+b} + 2^{a+b+c}.$$

Hence

$$F_\omega(x) = \frac{3^4 x + \left(3^3 + 3^2 2^a + 3\,2^{a+b} + 2^{a+b+c}\right)}{2^{a+b+c+d}},$$

so that

$$c(\omega) = 3^3 + 3^2 2^a + 3\,2^{a+b} + 2^{a+b+c}.$$

This illustrates concretely how the affine constant accumulates the propagated contributions of successive "+1" terms throughout the accelerated iteration.

B Continued Fractions, Tower Cycles, and the Extremal Envelope

This appendix records the Diophantine structure underlying the admissible grammar and explains the origin of the distinguished cycle lengths

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\ 31867,\ 79335,\dots$$

that appear throughout the paper.

B.1 Continued-Fraction Origin of the Tower Cycles

Let

$$\alpha = \log_2 3.$$

Its continued-fraction expansion begins

$$\alpha = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, \dots].$$

Writing

$$\frac{p_k}{q_k}$$

for the convergents of α, the first denominator sequence is

$$1,\ 1,\ 2,\ 5,\ 12,\ 41,\ 53,\ 306,\ 665,\ 15601,\dots$$

which coincides exactly with the tower lengths observed in the admissible grammar. The convergent denominators satisfy the standard recurrence

$$q_k = a_k q_{k-1} + q_{k-2},$$

where a_k is the corresponding continued-fraction coefficient. Consequently every tower cycle is assembled canonically from the two preceding tower cycles.

Table 1 records the first instances.

Table 1: Continued-fraction generation of the tower cycles.

q_k	Cycle	k	a_k	Cycle Composition	Sequence
1	C_0	0	–	Single	1
1	C_1	1	1	Double	2
2	C_2	2	1	$C_0 + 1 \cdot C_1$	$1, 2$
5	C_5	3	2	$2 \cdot C_2 + C_1$	$1, 2, 1, 2, 2$
12	C_{12}	4	2	$C_2 + 2 \cdot C_5$	$1, 2, 1, 2, 1, 2, 2, 1, 2, 1, 2, 2$
41	C_{41}	5	3	$3 \cdot C_{12} + C_5$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
53	C_{53}	6	1	$C_{12} + 1 \cdot C_{41}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
306	C_{306}	7	5	$5 \cdot C_{53} + C_{41}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
665	C_{665}	8	2	$C_{53} + 2 \cdot C_{306}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
15601	C_{15601}	9	23	$23 \cdot C_{665} + C_{306}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
31867	C_{31867}	10	2	$C_{665} + 2 \cdot C_{15601}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$
79335	C_{79335}	11	2	$2 \cdot C_{31867} + C_{15601}$	$1, 2, 1, 2, 1, 2, 2, \ldots, 1, 2, 1, 2, 2$

Thus the tower hierarchy is not an empirical feature of the grammar. It is forced by the Diophantine approximation structure of $\log_2 3$.

B.2 Alternating Expansion and Contraction

The convergents alternately overestimate and underestimate α.

Define

$$\varepsilon_k = q_k \alpha - p_k.$$

Then

$$\varepsilon_k \to 0$$

and the signs alternate.

Since

$$\alpha = \log_2 3,$$

it follows that

$$\varepsilon_k = q_k \log_2 3 - p_k = -\log_2 \left(\frac{2^{p_k}}{3^{q_k}} \right).$$

Equivalently,

$$\frac{2^{p_k}}{3^{q_k}} = 2^{-\varepsilon_k}.$$

Hence

$$\left|\log\left(\frac{2^{p_k}}{3^{q_k}}\right)\right| = (\log 2)\,|\varepsilon_k| \to 0.$$

Therefore

$$\frac{2^{p_k}}{3^{q_k}} \to 1.$$

The convergents alternately overestimate and underestimate $\log_2 3$, so the ratios alternately lie above and below unity while approaching exact multiplicative balance.

Table 2: Convergents of $\log_2 3$ and resulting dyadic–ternary synchronization ratios.

q_k	p_k	$\varepsilon_k = q_k \log_2(3) - p_k$	$2^{p_k}/3^{q_k}$
1	2	$-0.41594\ldots$	$1.33333\ldots$
2	3	$+0.16993\ldots$	$0.88889\ldots$
5	8	$-0.07519\ldots$	$1.05350\ldots$
12	19	$+0.01955\ldots$	$0.98654\ldots$
41	65	$-0.01654\ldots$	$1.01153\ldots$
53	84	$+0.00301\ldots$	$0.99791\ldots$

B.3 Primitive Return Words and the Failure of the Double Process

The Sturmian word generated by

$$d(n) = \lfloor n\alpha \rfloor + 1$$

admits a return-word decomposition. At the first nontrivial level, the primitive return words have lengths 2, 5, and 12. These correspond respectively to:

$$C_2 = (1, 2).$$

$$C_5 = (1, 2, 1, 2, 2).$$

$$C_{12} = (1, 2, 1, 2, 1, 2, 2, 1, 2, 1, 2, 2).$$

The 2-cycle is locally contractive, while the 5-cycle is locally expansive. A 12-cycle is formed by concatenating the 2-cycle with a pair of consecutive 5-cycles, but the 12-cycle is overall contractive.

One might ask why the extremal comparison process is based on the 5-cycle rather than the shorter expansive 1-cycle corresponding to a double increment.

Indeed, the shortest expansive block is

$$C_1 = (2),$$

for which

$$\frac{2^2}{3} = \frac{4}{3} > 1.$$

Repeated doubles generate the row sums

$$1, 2, 5, 14, 42, 132, 429, \ldots$$

which are the Catalan numbers. Since

$$C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}}$$

implies

$$\lim_{n\to\infty} \frac{C_{n+1}}{C_n} = 4.$$

Thus an infinite concatenation of doubles produces exponential growth rate 4, which exceeds the critical value 3.

The pure-double process is therefore unsuitable as a comparison envelope: it overestimates growth by too large a margin and cannot yield the desired spectral bound.

B.4 The Extremal Envelope

The main text proves that every admissible Sturmian word decomposes uniquely into primitive return words. Since the normalized cocycle contribution of C_5 exceeds that of every other primitive return word, replacing each return word by C_5 can only increase the average cocycle drift. Although the periodic comparison word C_5^∞ is itself forbidden by the Sturmian grammar, it provides a universal comparison envelope. The explicit evaluation of this envelope is carried out in Appendix C.

C Extremal Local Growth Calculations

This appendix provides the explicit finite computations underlying the local growth bounds used in the density argument. Recall that the sequence A020914 has first differences given by A022921, consisting of repeated blocks of *single* steps (s) and *double* steps (d).

C.1 Illustrative Extremal Computations

The generating rows evolve by a horizontal-sum rule:

$$a_i(n) = a_i(n-1) + a_{i-1}(n), \quad i, n \in \mathbb{N}, \tag{C.1}$$

so that sequences of single steps exhibit ordinary Pascal–type growth. Double steps differ only in that the terminal entry of the row is duplicated, causing the final term to appear twice in the generating tuple while preserving monotonicity of the preceding entries.

Lemma C.1 (Pascal-Type Horizontal Growth)**.** *Let a row generated by the horizontal-sum construction have strictly positive entries with total sum S.*

After one horizontal-sum step without duplication, the resulting row has total

$$S_{\text{new}} = S + S_{\text{int}},$$

where S_{int} denotes the sum of the nonterminal entries of the previous row.

Consequently

$$S_{\text{new}} < 2S.$$

Proof. The terminal entry of the new row equals the total sum S of the previous row. Every nonterminal entry contributes exactly once to the new row through the horizontal-sum recurrence. Hence

$$S_{\text{new}} = S + S_{\text{int}}.$$

Since all entries are strictly positive,

$$S_{\text{int}} < S,$$

and therefore

$$S_{\text{new}} < 2S.$$

$\square$

Corollary C.2 (Controlled Growth Under Duplication)**.** *A subsequent duplication contributes at most one additional copy of the terminal entry, whose value is S.*

Hence

$$S_{\text{new}} < 2S + S = 3S.$$

These examples are chosen so that each step saturates the inequalities of Lemma C.1, thereby realizing extremal growth. Equalities shown below should be interpreted as extremal upper-bound realizations rather than identities valid for all rows.

The following row-by-row computations realize extremal growth for the admissible local patterns and together determine the sharpest uniform bounds used in Proposition 10.1. Each example illustrates how a local pattern can saturate its maximal growth factor under the horizontal-sum dynamics. Let S denote the total sum of entries in the generating row prior to a given local pattern.

Example 1.

$$\textit{Single } a(10) = \underbrace{173}_{\text{sum}:=S} : \underbrace{1, 7, 25, 55, 85}_{S}$$

$$\textit{Double } a(11) = \underbrace{476}_{\text{sum}=3S} : \underbrace{1, 8, 33, 88}_{S} \underbrace{173}_{S}, \underbrace{173}_{S}$$

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum}=9S} : \underbrace{1, 9, 42, 130}_{2S} \underbrace{303}_{3S}, \underbrace{476}_{4S}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum}=33S} : \underbrace{1, 10, 52, 182}_{4S}, \underbrace{485}_{7S}, \underbrace{961}_{11S}, \underbrace{961}_{11S}$$

$$\textit{Double}\ a(14) = \underbrace{8045}_{\text{sum}=123S} : \underbrace{1, 11, 63, 245}_{8S}, \underbrace{730}_{15S}, \underbrace{1691}_{26S}, \underbrace{2652}_{37S}, \underbrace{2652}_{37S}$$

$$\textit{Single}\ a(15) = \underbrace{17637}_{\text{sum}=329S} : \underbrace{1, 12, 75, 320}_{16S}, \underbrace{1050}_{31S}, \underbrace{2741}_{57S}, \underbrace{5393}_{94S}, \underbrace{8045}_{131S}$$

The resulting maximal local ratios are

Pattern	Max Ratio
s, d	3/1
d, d	123/33
s, d, s	9/3
d, d, s	329/123

Example 2.

$$\textit{Single}\ a(12) = \underbrace{961}_{\text{sum:}=S} : \underbrace{1, 9, 42, 130, 303, 476}_{S}$$

$$\textit{Double}\ a(13) = \underbrace{2652}_{\text{sum}=3S} : \underbrace{1, 10, 52, 182, 485}_{S}, \underbrace{961}_{S}, \underbrace{961}_{S}$$

$$\textit{Double}\ a(14) = \underbrace{8045}_{\text{sum}=13S} : \underbrace{1, 11, 63, 245, 730}_{2S}, \underbrace{1691}_{3S}, \underbrace{2652}_{4S}, \underbrace{2652}_{4S}$$

$$\textit{Single}\ a(15) = \underbrace{17637}_{\text{sum}=37S} : \underbrace{1, 12, 75, 320, 1050}_{4S}, \underbrace{2741}_{7S}, \underbrace{5393}_{11S}, \underbrace{8045}_{15S}$$

$$\textit{Double}\ a(16) = \underbrace{51033}_{\text{sum}=131S} : \underbrace{1, 13, 88, 408, 1458}_{8S}, \underbrace{4199}_{15S}, \underbrace{9592}_{26S}, \underbrace{17637}_{41S}, \underbrace{17637}_{41S}$$

$$\textit{Single}\ a(17) = \underbrace{108950}_{\text{sum}=341S} : \underbrace{1, 14, 102, 510, 1968}_{16S}, \underbrace{6167}_{31S}, \underbrace{15759}_{57S}, \underbrace{33396}_{98S}, \underbrace{51033}_{139S}$$

The corresponding maximal local ratios are

Pattern	Max Ratio
s, d	3/1
d, d	13/3
s, d, s	341/131
d, d, s	37/13

Improved Extremal Alignment

Because the total multiplicative growth over a cycle is the product of the local growth factors, the geometric mean over one full cycle is maximized by selecting the maximal admissible factor at each local pattern occurrence. Consequently, once a uniform upper bound has been established for each admissible local pattern, the extremal global growth rate over a complete cycle is obtained by multiplying these maximal local factors according to their frequencies and taking the corresponding geometric mean.

Equivalently, because total growth over any admissible word is multiplicative in its local factors, the asymptotic per-symbol growth rate is the geometric mean of these factors; hence the maximal rate over the entire language generated by the grammar is achieved by the word that maximizes this mean subject to the grammar constraints.

Each example yields a valid (not necessarily sharp) upper bound on the maximal growth factor associated with each admissible local pattern. Since these bounds arise from the same uniform recurrence and apply globally to all admissible configurations, the true maximal growth factor for a given pattern is bounded above by the smallest recorded bound among all such valid upper bounds.

Because the recurrence is monotone with respect to the placement of delayed doubling, no other admissible configuration can exceed these extremal realizations. Accordingly, combining the sharpest bounds obtained across different initiating rows yields the tightest global envelope on admissible local growth.

Pattern	Max Ratio	Associated coefficient count
s, d	3	2
d, d	123/33	1
s, d, s	341/131	1
d, d, s	329/123	1

Consequently, the maximal geometric growth rate of a 5–cycle satisfies

$$\Big(3^2 \cdot (123/33) \cdot (341/131) \cdot (329/123)\Big)^{1/5} \approx 2.976 < 3. \tag{C.2}$$

Conclusion. Every admissible growth pattern of A186009 is constrained by the grammar of A022921. Infinite concatenations of 5-cycles, though forbidden, saturate local growth constraints and provide a strict upper envelope. Equation (C.2) confirms that even this extremal envelope has geometric growth rate < 3, ensuring subdominance relative to the denominator $2^{B(i)}$.

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